

Radionuclide Identification for Plastic Scintillation Detectors by Backward Cumulative Channel Sum

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Abstract

Large-area plastic scintillation detectors are widely used in radiation portal monitors, but their Compton-dominated pulse-height spectra provide limited radionuclide-specific information. We evaluated a backward cumulative channel sum (BCCS) representation combined with non-negative least-squares (NNLS) decomposition and compared it with two conventional baselines: NNLS fitting of the original spectrum with an explicit background template and detector-specific five-window NNLS. Nominal 20-s measurements were converted to count rate using the file-specific MCA live time. Background, twelve single-source spectra, and nine mixed-source spectra of Am-241, Cs-137, and Co-60 were acquired with a large-area plastic-scintillator portal-monitor detector. Samples 1, 6, and 10 served as reference responses, leaving eighteen known-composition spectra for independent evaluation. Under the stated known-**K** ranking criterion, all three methods returned the expected constituent set for all eighteen test spectra. For the nine mixed-source spectra, method-specific relative residuals were 0.48-1.31% for BCCS-NNLS, 0.88-1.60% for raw-spectrum NNLS, and 0.46-2.57% for five-window NNLS. Among the nine test spectra with net count rates at least 20% of background, all nine BCCS residuals were below 2%, compared with seven of nine raw-spectrum residuals and six of nine five-window residuals. Parametric Poisson resampling supported stable constituent ranking but did not include reference repositioning, gain drift, or environmental background changes. These findings establish BCCS-NNLS as a simple, linear, and interpretable representation for static, unshielded measurements under the tested conditions, while not constituting an open-set identification accuracy or a formal ISO 11929 detection-limit assessment.

Keywords: plastic scintillator; radiation portal monitor; radionuclide identification; backward cumulative channel sum; non-negative least squares; energy window; background subtraction

Introduction

Radiation portal monitors (RPMs) are deployed at border crossings, ports, transportation facilities, and industrial sites to detect unauthorized movement of radioactive material. Large polyvinyl-toluene or related plastic scintillation panels are attractive because they provide large sensitive areas, rapid response, mechanical robustness, and moderate cost [1,2]. Their principal spectrometric limitation is poor energy resolution: the measured pulse-height distribution is dominated by broad Compton continua rather than narrow full-energy peaks. Radionuclide identification must therefore use distributed spectral-shape information against a background that may exceed the net source response.

Several approaches have been developed for low-resolution plastic spectra. Energy-window algorithms compress the spectrum into a small number of detector-specific regions and use integrated counts or compensated ratios [3,4]. They are computationally efficient, but the selected boundaries depend on gain, detector response, and the target library, and within-window information is discarded. Raw-spectrum least-squares fitting retains the individual channels and can include background explicitly, but local counting

fluctuations and the dominance of the background component can complicate interpretation. Energy-weighted, principal-component, inverse-calibration, and machine-learning methods provide additional alternatives [5-11], often at the cost of detector-specific calibration, training data, or more elaborate decision rules.

The Backward Cumulative Channel Sum (BCCS) considered here reports, at each threshold channel, the count integrated from that threshold to the highest acquired channel. The transformation visually averages local channel fluctuations while retaining a continuous sequence of threshold-dependent spectral responses. It is also linear: the BCCS of a sum of spectra equals the sum of their BCCS representations. This property permits direct non-negative linear decomposition of mixed spectra using fixed single-radionuclide reference responses.

The initial study demonstrated qualitative separability but lacked sufficient quantitative validation, experimental detail, and comparison with established approaches. The revised study therefore (i) specifies the detector, acquisition, source geometry, reference selection, and channel treatment; (ii) uses file-specific live-time normalization and a quantitative residual; (iii) compares BCCS-NNLS with raw-spectrum NNLS and detector-specific five-window NNLS; (iv) reports all single- and mixed-source results rather than selected examples alone; and (v) evaluates counting-statistical stability by parametric Poisson resampling.

The primary objective is not to claim universal or open-set identification accuracy. Instead, we assess whether BCCS provides a practical linear representation for known-library decomposition under the static, unshielded measurement conditions examined here, and identify the signal-strength and goodness-of-fit conditions under which the results are most stable.

Results

Reference responses and distance dependence

The reference and distance-series measurements retained the broad continuum responses expected from a plastic scintillation detector. After channel-level background subtraction and normalization to the selected reference distance for each radionuclide, the measured response decreased consistently with increasing source-to-detector distance (Table 1). The known radionuclide remained top-ranked in all single-source measurements.

Parametric resampling showed 100% ranking stability for most measurements. Slightly lower values were obtained only for the weakest measurements: 96.86% for Am-241 at 460 cm, 99.84% for Cs-137 at 390 cm, and 99.84% for Co-60 at 470 cm. At these low response levels, the off-target coefficients and BCCS fitting residuals also increased. Therefore, correct top ranking alone should not be interpreted as sufficient evidence of reliable rejection of all non-target radionuclides.

The measured distance dependence did not follow the inverse-square relationship exactly, particularly at short source-to-detector distances. The inverse-square approximation assumes a point-like detector in the far field, whereas the detector used in this study had a large active area of approximately 30×180 cm² and subtended a substantial solid angle at short distances. The measured responses were therefore compared with finite rectangular-detector solid-angle estimates rather than with the $1/r^2$ approximation alone. The normalized measured response was calculated from the background-subtracted total count rate integrated over Channels 4–256 and normalized to the selected reference measurement for each radionuclide. The finite-detector estimates were calculated for a centered rectangular detector with an active area of 30×180 cm² and normalized to the same reference distances.

As shown in Table 1, the measured responses of Cs-137 and Co-60 were in close agreement with the finite-detector solid-angle estimates. The larger deviations observed for Am-241 may reflect its lower signal-to-

background ratio and the greater sensitivity of its low-energy response to the lower-level discriminator setting, background mismatch, source-positioning uncertainty, and scattering from surrounding materials.

Table 1. Comparison of normalized measured responses with inverse-square and finite rectangular-detector solid-angle estimates.

Radionuclide	Distance (cm)	Inverse-square estimate	Finite-detector solid-angle estimate	Normalized measured response	Top-rank stability (%)
Am-241	80	1.000	1.000	1.000	100.00
	180	0.198	0.269	0.316	100.00
	280	0.082	0.118	0.129	100.00
	460	0.030	0.045	0.042	96.86
Cs-137	70	3.449	2.545	2.581	100.00
	130	1.000	1.000	1.000	100.00
	300	0.188	0.220	0.220	100.00
	390	0.111	0.132	0.128	99.84
Co-60	60	4.000	2.728	2.763	100.00
	120	1.000	1.000	1.000	100.00
	310	0.150	0.181	0.183	100.00
	470	0.065	0.081	0.074	99.84

Responses were normalized to the measurements obtained at 80 cm for Am-241, 130 cm for Cs-137, and 120 cm for Co-60. The inverse-square estimates were calculated relative to the corresponding reference distance. The finite-detector estimates were calculated from the solid angle subtended by a centered rectangular detector with an active area of approximately 30×180 cm².

Mixed-source BCCS decomposition

BCCS-NNLS recovered the known constituent pair as the two largest coefficients in all eight binary mixtures. The ternary Sample-17 returned positive coefficients for all three reference components. For Sample-13 (Cs-137 + Co-60), the BCCS coefficients were 0.032 for Am-241, 0.949 for Cs-137, and 0.797 for Co-60, with a residual of 0.48%. The 95% counting-statistical intervals were 0.000-0.066, 0.924-0.974, and 0.775-0.820, respectively. For Sample-20 (Am-241 + Cs-137), the coefficients were 0.946, 0.609, and 0.019, with a residual of 0.87%; the corresponding 95% intervals were 0.913-0.979, 0.588-0.629, and 0.000-0.038.

Small positive coefficients for absent radionuclides were retained because they quantify spectral leakage, reference mismatch, and counting fluctuations. They should not be converted directly into activity fractions or automatically interpreted as additional radionuclides without validated radionuclide-specific decision thresholds.

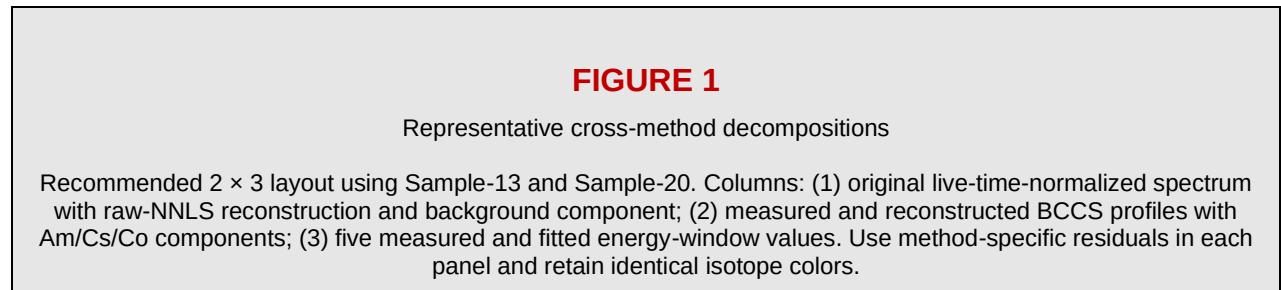


Figure 1. Representative raw-spectrum, BCCS, and five-window NNLS results for Sample-13 and Sample-20. Coefficients are detector-response scaling factors relative to Samples 1, 6, and 10 and do not represent activity fractions.

Comparison with raw-spectrum and five-window NNLS

Across the eighteen independent spectra with known compositions, BCCS-NNLS, raw-spectrum NNLS, and five-window NNLS all returned the expected single radionuclide or expected binary pair under the known-K ranking rule. Sample-17 was treated as a consistency case because all three library radionuclides were known to be present.

For the nine mixed-source spectra, BCCS residuals ranged from 0.48% to 1.31%, raw-spectrum residuals from 0.88% to 1.60%, and five-window residuals from 0.46% to 2.57%. BCCS produced a lower residual than five-window NNLS in seven of nine mixtures; the window representation was lower for Samples 15 and 20. Raw-spectrum residuals were below 2% for all mixed spectra. These numerical residuals are descriptive within their own representation spaces and are not directly interchangeable goodness-of-fit probabilities.

Table 2. Cross-method results for the nine mixed-source spectra. Residuals were calculated in the respective BCCS, original-channel, and five-window spaces and should not be interpreted as identical statistical quantities.

Sample	Expected constituents	Rnet (%)	BCCS residual (%)	Raw residual (%)	Window residual (%)	Known-K ranking
13	Cs-137 + Co-60	89.72	0.48	1.60	1.12	Correct for all three methods
14	Am-241 + Co-60	13.53	1.31	0.88	2.57	Correct for all three methods
15	Cs-137 + Co-60	16.85	1.20	0.99	1.02	Correct for all three methods
16	Am-241 + Cs-137	44.56	0.53	0.97	1.43	Correct for all three methods
17	Am-241 + Cs-137 + Co-60	24.68	1.31	1.16	2.41	All three components positive
18	Co-60 + Am-241	38.23	0.78	1.11	1.46	Correct for all three methods
19	Cs-137 + Co-60	20.45	1.12	0.96	1.14	Correct for all three methods
20	Am-241 + Cs-137	54.82	0.87	1.15	0.46	Correct for all three methods
21	Cs-137 + Co-60	29.65	1.30	1.11	1.62	Correct for all three methods

Signal strength and conservative operating region

The relationship between reconstruction stability and net signal was evaluated by grouping the eighteen independent spectra according to Rnet. BCCS residuals increased sharply for some spectra below 10% of background, particularly the weakest Am-241 measurements. All twelve spectra with Rnet at least 10% had BCCS residuals below 2% in the present data set. A more conservative 20% criterion was retained for practical interpretation.

Nine independent spectra had Rnet at least 20%. All three methods returned the correct known-K ranking for all nine. All nine BCCS residuals were below 2% (0.48-1.57%), compared with seven of nine raw-spectrum residuals and six of nine five-window residuals. The raw-spectrum result must be interpreted cautiously because background dominates the original channel-space norm; conversely, the BCCS and window residuals are calculated after background subtraction.

Table 3. Observed known-K ranking and residual results by net-to-background count-rate range. The sample numbers per bin and all fitted coefficients are provided in Supplementary Table S2.

Rnet range (%)	N	BCCS correct	BCCS $\epsilon < 2\%$	Raw correct	Raw $\epsilon < 2\%$	Window correct	Window $\epsilon < 2\%$
<5	3	3/3	0/3	3/3	3/3	3/3	0/3
5 to <10	3	3/3	1/3	3/3	3/3	3/3	2/3
10 to <20	3	3/3	3/3	3/3	3/3	3/3	1/3
≥ 20	9	9/9	9/9	9/9	7/9	9/9	6/9

FIGURE 2

Method-specific residual versus net-to-background count-rate ratio

Scatter plot with Rnet (%) on the x-axis and relative residual (%) on the y-axis. Plot BCCS, raw-spectrum, and five-window results with different markers; label sample numbers. Add a vertical dashed line at Rnet = 20% and a horizontal dashed line at residual = 2%. Consider a logarithmic x-axis to display the 1-135% range. Exclude references.

Figure 2. Dependence of method-specific reconstruction residual on net signal strength. Residuals are shown to compare stability trends within each representation and are not identical chi-square statistics across methods.

Counting-statistical stability

For the eight binary mixed-source spectra, the expected pair occupied the two largest BCCS coefficients in 100% of the 5,000 Poisson resamples. All three Sample-17 coefficients were positive in 100% of resamples. Single-source ranking stability remained at least 96.86%, with the lowest value occurring for the weakest Am-241 measurement. These results indicate strong stability against counting fluctuations around the observed spectra.

The result does not imply 100% field identification accuracy. The resamples do not reproduce independent source placement, background nonstationarity, temperature-dependent gain shifts, detector aging, cargo shielding, or radionuclides absent from the reference library. The empirical background-only coefficient gate and the Rnet operating region should therefore be used as quality-control indicators rather than universal decision thresholds.

Methods

Experimental system and measurement geometry

Measurements were performed using a cRPM-54 testbed equipped with a large-area plastic scintillation detector panel measuring $30 \times 180 \times 5 \text{ cm}^3$ and coupled to a photomultiplier tube. High-voltage supply and pulse-height acquisition were provided by a custom M1 International system incorporating an Amptek DP5G digital pulse processor [14]. The photomultiplier high voltage was set to 1,475 V, the amplifier gain was set to 4.0, and the acquisition system recorded 256 pulse-height channels.

The radioactive materials were sealed point sources with nominal activities of 90 μCi for Am-241, 16 μCi for Cs-137, and 9 μCi for Co-60. The front surface of the detector panel was defined as the source-to-detector distance origin. Each source was positioned on the central normal axis of the detector, with horizontal and vertical alignment referenced to the center of the active detector area. The source height was set to the detector-center height.

For Samples 13–17, all constituent sources were positioned at the same nominal source-to-detector distance. For Samples 18–21, the constituent sources were positioned at different distances while maintaining alignment with the detector center. No intentional shielding, cargo, or other intervening material was introduced.

These conditions represent static, unshielded measurements of sealed point sources. They do not reproduce vehicle occupancy, cargo attenuation, complex load-dependent scattering, moving-source profiles, or other operational conditions encountered in portal monitoring. The results should therefore be interpreted as a controlled feasibility study specific to the detector configuration and reference library used in this work.

The experimental arrangement and the reference responses used to construct the NNLS library are shown in Fig. 3. Each reference measurement was normalized using its file-specific live time before subsequent spectral processing.

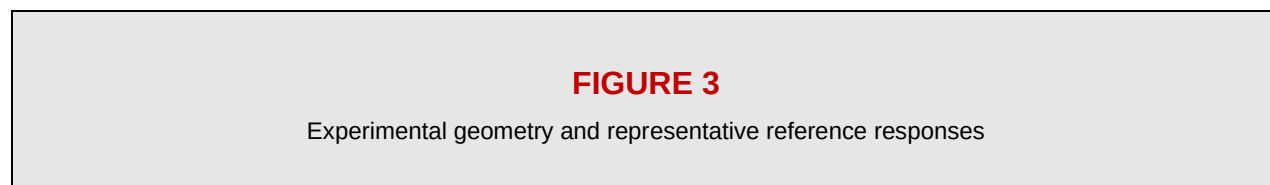


Figure 3. Experimental configuration and reference responses.

(a) Photograph of the cRPM-54 detector panel and sealed point-source positioning arrangement. (b) Schematic front and side views showing the detector front surface as the source-to-detector distance origin and the sealed point source aligned with the horizontal and vertical center of the active detector area. (c–e) Background-subtracted reference spectra of Am-241, Cs-137, and Co-60 used to construct the NNLS reference library. Each spectrum was normalized using its file-specific live time.

Data set, acquisition time, and channel treatment

The completed data set comprised one independently acquired background spectrum, twelve single-source spectra (Samples 1-12), and nine mixed-source spectra (Samples 13-21). The complete conditions are listed in Supplementary Table S1. Sample-22 was not included in the formal analysis because its source identity and geometry could not be verified.

All acquisitions had a nominal real-time duration of 20 s. The actual MCA live time recorded in the file headers ranged from 18.672 to 20.000 s; therefore, each channel was divided by its file-specific live time before comparison. The background live time was 20.000 s. The background contained 121,966 counts over Channels 4-256, corresponding to $6,098.3 \text{ s}^{-1}$.

Channels 1-3 were below the configured lower-level discriminator and contained no accepted events; Channels 4-256 were treated as the valid acquisition range. The revised primary BCCS analysis used the complete valid range and did not truncate the spectrum to an undefined 'low-channel region'. Differences among the reference BCCS curves were largest at lower threshold channels, but every BCCS coordinate retained all counts above its threshold.

No traceable channel-to-energy calibration was established for this data set. Because the plastic-scintillator spectra did not provide narrow photopeaks suitable for conventional centroid and FWHM calibration, the analysis was performed in detector-channel space. This choice avoids assigning unsupported physical energies but makes the reference library and five-window boundaries specific to the present gain, discriminator, and channel configuration. Recalibration is required after gain, LLD, or electronics changes.

Sample-1 (Am-241 at 80 cm), Sample-6 (Cs-137 at 130 cm), and Sample-10 (Co-60 at 120 cm) were used as fixed single-radionuclide reference spectra. The remaining eighteen known-composition spectra were treated as independent tests. The fitted coefficients are scaling factors relative to these detector responses; they are not activity fractions because source activities, distances, geometry, attenuation, and energy-dependent efficiency differ.

Backward cumulative channel sum and background compensation

Let C_i denote the live-time-normalized count rate in channel i and let $N = 256$. The BCCS at threshold i is defined as

$$B_i = \sum_{j=i}^N C_j \quad (1)$$

For an additive mixture, the same linear operator T satisfies $T(C_1 + C_2) = T(C_1) + T(C_2)$. Background correction was therefore performed without breaking linearity:

$$S^M = T(C^M) - T(C^{bg}) = T(C^M - C^{bg}) \quad (2)$$

where C^M and C^{bg} are the live-time-normalized measurement and background spectra. Negative channel-level net values arising from counting fluctuations were retained before the BCCS transformation. The primary BCCS measurement vector contained all threshold coordinates from Channel 4 through Channel 256.

A background-ratio representation, $T(C^M)/T(C^{bg}) - 1$, may improve visual contrast when the background is stable. However, denominator uncertainty can amplify weak high-channel fluctuations. It was therefore retained only as a secondary visualization and was not used for fitted coefficients, residuals, ranking statistics, or signal-strength conclusions.

NNLS decomposition, ranking rule, and residual

The background-subtracted BCCS vector y was modeled using a reference matrix A whose columns were the Am-241, Cs-137, and Co-60 BCCS reference profiles. The coefficient vector was estimated by NNLS [12]:

$$\hat{a} = \arg \min_{a \geq 0} \|y - Aa\|_2^2 \quad (3)$$

The method-specific relative reconstruction residual was

$$\varepsilon = 100 \times \|y - A\hat{a}\|_2 / \|y\|_2 \quad (4)$$

For a single-source spectrum, the known radionuclide was considered correctly ranked when it had the largest coefficient. For a binary mixture, the two known constituents were required to occupy the two largest coefficient ranks. This is a known-K evaluation because the number of expected constituents is supplied in advance. For the ternary Sample-17, $K = 3$ equals the size of the available radionuclide library; positive recovery of all three coefficients was therefore treated as decomposition consistency rather than a discriminative top-K test. No open-set radionuclide-specific decision thresholds were validated.

Adjacent BCCS thresholds share counts and therefore have correlated uncertainties. The unweighted BCCS residual describes reconstruction agreement within cumulative space but is not a chi-square probability and should not be interpreted as independent evidence at every threshold.

Raw-spectrum NNLS baseline

Following conventional comparison, the original live-time-normalized spectrum was fitted over Channels 4-256 without subtracting the background from the measurement vector. The reference matrix contained the measured background template b and the three net single-radionuclide responses s_x :

$$y^{\text{raw}} = \beta b + a_{\text{Am}} s_{\text{Am}} + a_{\text{Cs}} s_{\text{Cs}} + a_{\text{Co}} s_{\text{Co}} + e \quad (5)$$

All coefficients were constrained to be non-negative. The fitted background coefficient β was excluded from radionuclide ranking. The residual was calculated in original channel space. Because the gross spectrum is dominated by background at weak signal levels, a small raw-spectrum residual can principally reflect an accurate background fit and does not alone establish radionuclide discrimination.

Detector-specific five-window NNLS baseline

A low-dimensional energy-window baseline was implemented using the established principle that PVT window boundaries must be calibrated for the detector and target source set rather than copied from another RPM [3,4]. The window limits were selected from the present single-radionuclide reference responses and fixed before evaluation of the test spectra.

Window	Channel range	Number of channels	Primary role
W1	4-10	7	Am-241-dominant low-channel region
W2	11-25	15	lower Cs-137-sensitive region
W3	26-40	15	upper Cs-137-sensitive / transition region
W4	41-80	40	Co-60-sensitive region
W5	81-256	176	High-channel control region sensitive to residual background mismatch

Table 4. Detector-specific energy windows used for the five-window NNLS baseline.

After the same live-time normalization and channel-by-channel background subtraction used for BCCS, the net count rate was integrated within each window to form a five-element vector. The corresponding five-element Am-241, Cs-137, and Co-60 reference responses were decomposed by NNLS. Because an independently measured background spectrum was subtracted before window integration, W5 was not used to estimate a separate background coefficient. Instead, it served as a high-channel control interval sensitive to residual background mismatch and high-energy spectral contributions.

Signal-strength measure and empirical operating region

Signal strength relative to the independently measured background was expressed as

$$R_{\text{net}} = 100 \times (C_{\text{sample}} - C_{\text{background}}) / C_{\text{background}} \quad (6)$$

where the count rates were integrated over Channels 4-256. Thus, $R_{\text{net}} = 20\%$ corresponds to a gross sample count rate equal to 120% of the background rate. The 20% level was evaluated as a conservative empirical operating region; it was not selected as a universal decision threshold or detection limit.

Parametric counting-statistical resampling

Counting-statistical stability was examined using 5,000 parametric Poisson resamples per independent test spectrum. Each measurement channel and background channel was independently sampled using its observed count as the Poisson mean, converted to count rate with the original file live time, background-subtracted, transformed by BCCS, and fitted against the fixed measured reference matrix. The coefficient distributions were summarized by the median and 2.5th-97.5th percentiles.

Ranking stability was defined as the fraction of resamples returning the known single radionuclide or known binary pair in the required top-K positions. For Sample-17, the fraction with all three coefficients positive was reported. This is an empirical model-based screening gate, not a formal characteristic limit. ISO 11929-3:2025 requires a defined measurand, full uncertainty model, covariance treatment, and specified error probabilities for unfolding-based decision thresholds and detection limits [13].

The resampling treats the experimentally measured reference matrix as fixed. It therefore quantifies counting variation around the observed spectra but does not include uncertainty from independent reference acquisition, source repositioning, activity certification, detector gain drift, aging, temperature, or environmental background changes.

Discussion

Interpretation of the method comparison

The three representations preserve different information. Raw-spectrum NNLS retains every original channel and explicitly models the background, but the residual can remain small at weak signal levels because the fitted background template dominates the norm. Five-window NNLS compresses the net spectrum to five robust values and successfully retained the known constituent ranking in the present data, but its performance depends on detector-specific boundaries and it discards variation within each window. BCCS retains a value at every threshold while integrating all counts above that threshold, providing a continuous linear representation between the original spectrum and a small-window compression.

The BCCS transformation should not be described as creating independent additional information. Adjacent cumulative coordinates are strongly correlated. Its practical benefit is representational: broad source-dependent changes are expressed as smooth threshold-response profiles that are compatible with linear NNLS decomposition. A future generalized least-squares treatment could incorporate the cumulative covariance matrix rather than using an unweighted residual.

The energy-window baseline was intentionally detector-specific. The ranges reported for other RPMs were not transferred directly because their channel-to-energy relation, detector response, gain, and calibration sources differ [3,4]. The current window boundaries must likewise be recalculated after changes in gain, discriminator, or target library. BCCS reduces dependence on a small number of boundaries but remains dependent on stable channel registration and representative references.

Interpretation of coefficients, residuals, and the 20% criterion

The NNLS coefficients quantify the scale of each fixed detector-response template required to reconstruct a measurement. A coefficient of one means a response comparable in magnitude to the corresponding reference measurement after the stated preprocessing. It does not imply equal activity, equal photon emission rate, or equal activity fraction. Quantitative activity estimation would require traceable source activities and a geometry- and energy-dependent detector-response model.

The observed 18/18 known-K ranking is a result for the present finite data set, not a general accuracy estimate. The 20% net-to-background level was selected as a conservative empirical operating region because all nine qualifying BCCS spectra had residuals below 2%, even though the present data also performed well at 10-20%. A count-rate criterion alone cannot establish library consistency. A measurement may exceed the background by more than 20% and still be inconsistent with the available templates because of an unknown radionuclide, gain shift, altered geometry, or changed background. Acceptance should therefore combine signal strength, residual or other goodness-of-fit information, and a validated radionuclide decision rule.

The empirical Rnet and residual criteria are not formal ISO 11929 characteristic limits. A standard-based evaluation for an unfolding problem would require a defined radionuclide measurand, uncertainty propagation for measurement, background, reference response, calibration and geometry, correlations among channels, and predefined decision error probabilities [13].

Practical scope and limitations

Although the present study used static acquisitions, the BCCS transformation is not restricted to offline analysis. The transformation and subsequent NNLS decomposition can be applied sequentially to spectra acquired over short, fixed-duration time windows. For a localized source passing a portal detector, the source contribution is expected to increase during the interval near closest approach and to return toward the background level before and after passage. The pre- and post-passage intervals may therefore provide contemporaneous background information, whereas the transient high-count interval can be used for radionuclide ranking and decomposition. This processing structure makes the BCCS method compatible with real-time portal-monitoring applications.

Moving-source measurements and quantitative processing-latency tests were not included in the dataset analyzed in the present study. Therefore, although the method is computationally applicable to sequential short-duration spectra, its real-time detection probability, alarm latency, false-alarm rate, and radionuclide-identification performance under moving-source conditions were not quantitatively evaluated here.

The present measurements used static, unshielded sealed point sources aligned with one large-area detector panel. The study did not include vehicle occupancy, extended sources, cargo attenuation, source shielding, complex load-dependent scattering, moving-source profiles, or changing environmental backgrounds. The results should therefore be interpreted as a controlled feasibility study for the detector configuration and reference library used.

Each experimental condition was represented by a single acquisition. Parametric resampling increased the number of counting-statistical realizations but could not replace independent repeated measurements. The reference spectra were treated as fixed, and uncertainties in source activity and reference measurements were not propagated. In addition, the reference library contained only Am-241, Cs-137, and Co-60. The ranking analysis was therefore closed-library and known-K; it did not establish false-positive rates for absent library members, interfering radionuclides, or unknown sources.

The absence of a traceable energy calibration also limits direct transferability to other detector systems. Although channel-based operation is a practical advantage for a fixed detector configuration, changes in

detector gain, lower-level discriminator setting, temperature, or detector aging may shift the spectral response and invalidate both the BCCS reference profiles and the fixed energy-window definitions. Routine reference measurements, background monitoring, and gain-stability acceptance criteria would therefore be required before operational deployment.

Further validation should include independent repeated measurements, measurements on multiple background days, controlled temperature and gain perturbations, source repositioning, shielding and representative cargo materials, additional radionuclides and naturally occurring radioactive materials, and open-set decision thresholds. Moving-source studies should additionally evaluate acquisition-window duration, source speed, closest approach, transient detection probability, correct-identification duration, alarm latency, and false-alarm rate. Generalized least-squares or uncertainty-aware Bayesian decomposition may provide a more rigorous treatment of cumulative-channel covariance and reference-spectrum uncertainty.

Conclusion

BCCS combined with NNLS provided a linear and computationally simple representation for decomposing Am-241, Cs-137, and Co-60 responses measured with a large-area plastic-scintillator RPM detector. Using live-time-normalized spectra, a fixed three-reference library, and full Channels 4-256, BCCS, raw-spectrum, and five-window NNLS all returned the correct known-K constituent ranking for the eighteen independent known-composition tests. BCCS produced residuals below 2% for all nine spectra with net count rates at least 20% of background and for all nine mixed-source spectra.

These results support practical feasibility under the tested static, unshielded conditions. They do not demonstrate open-set identification accuracy, activity quantification, or a universal count-rate threshold. Operational use requires a representative and stable reference library, background and gain monitoring, method-specific goodness-of-fit criteria, and validation under realistic cargo and environmental conditions.

Data availability

The raw MCA spectra, measurement-condition table, processed spectra, fitted coefficients, resampling summaries, and analysis tables supporting this study will be deposited in a DOI-assigning public repository before resubmission. Repository and DOI: [ZENODO DOI TO BE INSERTED].

AUTHOR ACTION REQUIRED: Deposit the complete raw and processed data package in Zenodo and replace the placeholder above with the DOI.

Code availability

The code used for live-time normalization, background correction, BCCS transformation, raw-spectrum NNLS, five-window NNLS, parametric Poisson resampling, and generation of the reported tables and figures will be deposited in the same DOI-assigning repository. The archive should include source code, dependency information, fixed random seeds, parameter settings, and reproduction instructions. Repository and DOI: [ZENODO DOI TO BE INSERTED].

AUTHOR ACTION REQUIRED: The Editor explicitly requires a DOI-assigned code repository. This placeholder must be completed before submission.

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Author contributions

M.M. conceived and supervised the study, performed the experiments, and served as the corresponding author. J.-Y.S. developed the analysis software and performed the signal processing. B.J. organized the test data and supported the evaluation. All authors contributed to interpretation of the results and reviewed the manuscript.

Competing interests

The authors declare no competing interests.

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